

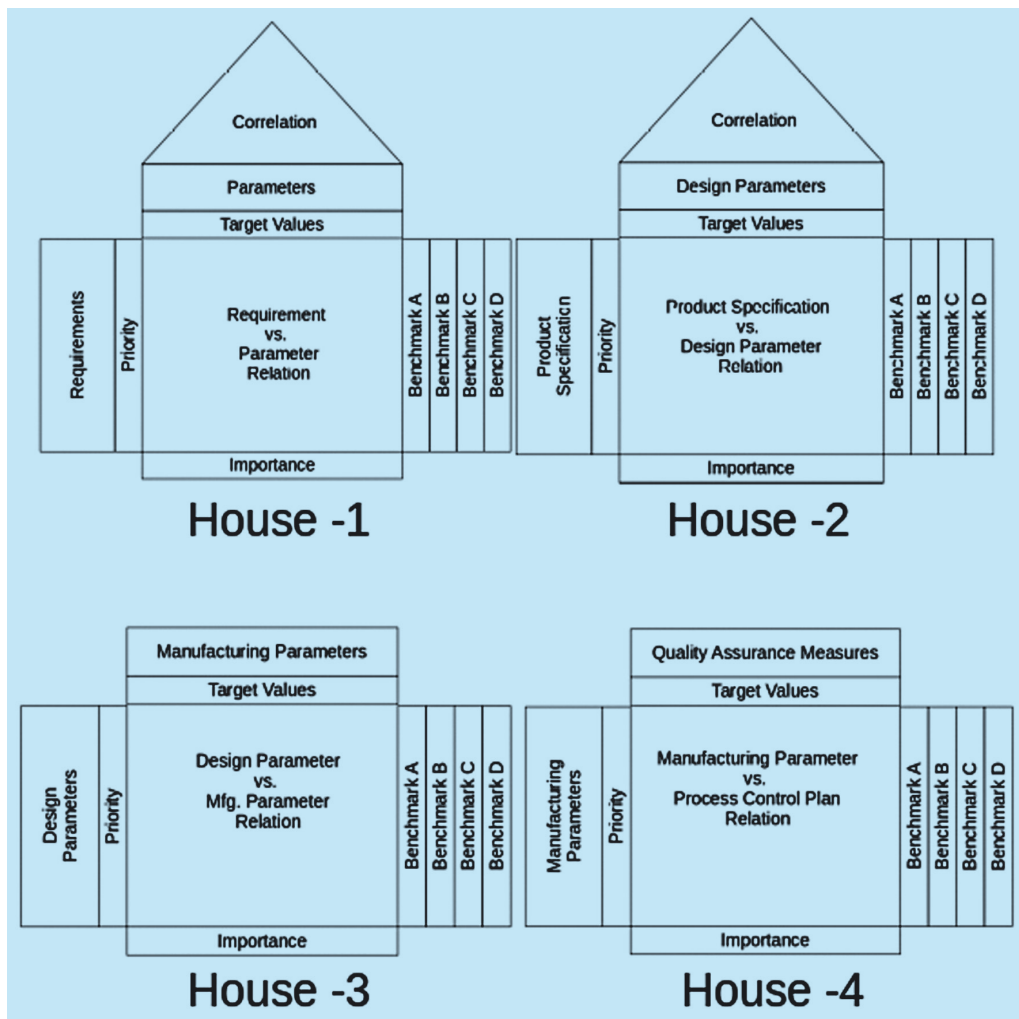


Fig. 2: Full set of QFD charts

important. Some of the requirements may be just a wish, or what we call nice to have features. Other requirements may be strongly desired and may influence the buying decisions. The third category of requirements is the must-have features. Unless the product has those must-have features it is not usable.

We rate these priorities on a scale of 1 to 10, with 1 being for the least essential and 10 for the most important requirements. Typically, ratings 7 to 10 are reserved for requirements that are necessary for the product. Customers will not purchase the product if these requirements are not fulfilled. The next set of priorities is what sets one product

apart from the other. These requirements get a rating between 4 and 7.

The rest of the requirements are what customers will appreciate but will not strongly motivate or influence them to purchase. Such requirements typically get a rating between 1 and 4. Understanding the priority of the requirements is as important as understanding the requirements.

Once we get a fair idea of the requirements and their priority, we should look at the existing products and benchmark and select the combinations of features for the product that we want to offer.

All these features and combinations can be quite confusing and

difficult to handle. The QFD process provides a concise and systematic way to handle this huge amount of information. For QFD we create a chart to capture product requirement information. In the first two columns we write down the requirements and their priority. The product needs to have a certain feature to address the requirements.

For example, if we say our charge controller should handle 600W of power at 24V, then the transistor or relay that will divert the power has to be rated for that current and voltage. One feature may address multiple requirements, or one requirement may get addressed by multiple features.

We write down these product features in columns, starting from the third column.

We put the number 1 to indicate which of the requirement is addressed by which product feature. At end of the exercise, we should have mapped all the product requirements to product features. After the product feature columns, we write the benchmark products and rate them on performance against the product requirements.

This exercise gives us clarity on which product features address which requirements and the performance of other products against various requirements. In addition to the captured information, we calculate the relative importance of the product features by taking the weighted sum of the product of